

Iris Dataset Documentation

Comprehensive README & Dataset Overview

1. Project Overview

This document serves as the formal documentation for the Iris Flower Classification project. The analysis explores the morphometric variation of three iris species (Setosa, Versicolor, and Virginica) based on sepal and petal measurements.

Samples: 150

Features: 4

Classes: 3

Status: Cleaned

2. Dataset Schema

Feature	Description	Type
sepal_length	Length of the sepal in cm	Numerical (float)
sepal_width	Width of the sepal in cm	Numerical (float)
petal_length	Length of the petal in cm	Numerical (float)
petal_width	Width of the petal in cm	Numerical (float)
species	Flower class (Setosa, Versicolor, Virginica)	Categorical

3. Summary Statistics

The following table presents a statistical snapshot of the 150 observations analyzed in the `Iris` dataset. `ipynb` notebook.

Stat	Sepal Length	Sepal Width	Petal Length	Petal Width
count	150.00	150.00	150.00	150.00
mean	5.84	3.06	3.76	1.20
std	0.83	0.44	1.77	0.76
min	4.30	2.00	1.00	0.10
25%	5.10	2.80	1.60	0.30
50%	5.80	3.00	4.35	1.30
75%	6.40	3.30	5.10	1.80
max	7.90	4.40	6.90	2.50

4. Key Insights

- **Linear Separability:** Iris Setosa is highly distinct and can be easily classified using simple linear thresholds, particularly with petal length.
- **Feature Importance:** Petal dimensions (length and width) show much higher variance between species than sepal dimensions, making them more significant for machine learning models.
- **Data Quality:** No missing values were detected, ensuring high reliability for statistical modeling.

5. Technical Implementation

The associated notebook utilizes the following tech stack:

- **Pandas:** For data manipulation and structural analysis.
- **Seaborn/Matplotlib:** For high-level statistical data visualization and pair-plotting.